

A Machine Learning Framework for Day-Ahead Electricity Price Forecasting with Robust Trading Strategy Evaluation and Production-Ready Validation

Energy Demand Forecast Project
<https://github.com/rav-lad/energy-demand-forecast>

November 19, 2025

Executive Summary

Objective. This study develops a complete machine learning framework for day-ahead electricity price forecasting on the French market, and evaluates the downstream impact on systematic trading strategies. The objective is to deliver a pipeline suitable for production, from data collection to forecasting, backtesting, validation, and model monitoring.

Methods. We evaluate five predictive models (Random Forest, XGBoost, LightGBM, Ridge Regression, GRU) over 700 days of historical data (2023–2024). The framework includes:

- feature engineering (lags, rolling statistics, meteorological data, calendar effects),
- strict leakage audits,
- realistic trading simulation with transaction costs,
- Sharpe ratio correction,
- robustness checks and extreme-event analysis.

Key Findings.

- XGBoost achieves the strongest predictive performance ($R^2 = 0.686$, MAPE $\approx 30\%$).
- Random Forest yields the best trading performance (Sharpe 1.65, annualized return 88%, max drawdown -4.2%).
- Tree-based models outperform the GRU due to the tabular, low-sample nature of the dataset.
- A critical Sharpe annualization bug inflated results by 62%; the corrected methodology aligns with institutional standards.

Production Readiness. A robust pipeline is achieved, with no detected data leakage. For deployment, the following enhancements are recommended:

- replacing historical weather observations with forecasts,

- adding walk-forward validation,
- implementing drift monitoring and automated retraining triggers.

Conclusion. This framework demonstrates that high-quality electricity price forecasts and profitable trading strategies are attainable using non-deep-learning models, provided the pipeline follows rigorous validation and error controls.

Abstract

This study presents a complete machine learning framework for forecasting day-ahead electricity prices in the French market and evaluates its application in systematic trading strategies. Five models are benchmarked—Random Forest, XGBoost, LightGBM, Ridge Regression, and a compact GRU network—using 700 days of historical data. XGBoost delivers the highest predictive accuracy ($R^2 = 0.686$), while Random Forest achieves superior risk-adjusted trading performance (Sharpe 1.65, annualized return 88%). A thorough data-leakage audit and transaction-cost modeling are conducted, and a critical Sharpe annualization bug inflating results by 62% is corrected. The study provides an interpretable, reproducible, and production-ready methodology for ML-based electricity trading systems.

Contents

1	Introduction	5
1.1	Research Objectives	5
1.2	Contributions	5
2	Data Description	5
2.1	Sources	5
2.2	Descriptive Statistics	6
2.3	Data Cleaning	6
3	Exploratory Data Analysis	6
3.1	Time Series Structure	6
3.2	Distributional Properties	7
3.3	Correlation Structure	8
4	Feature Engineering	8
4.1	Temporal Features	8
4.2	Lag Features	8
4.3	Rolling Statistics	9
4.4	Total Feature Set	9
5	Methodology	9
5.1	Train–Test Split	9
5.2	Models Evaluated	9
5.3	Walk-Forward Recommendation	9
6	Model Benchmark Results	10
6.1	Interpretation	10
6.2	Feature Importance (SHAP)	11
7	Trading Strategy Evaluation	12
7.1	Summary Performance	12
7.2	Why RF Trades Best	12
7.3	Equity Curve Visualization	12
7.4	Risk–Return Profile	13
7.5	Extreme Event Analysis	14
8	Discussion and Limitations	14
8.1	Realistic Performance Expectations	14

8.2	Production Considerations	15
8.3	Limitations	15
9	Conclusion	15
10	Appendix	16

1 Introduction

Day-ahead electricity price forecasting plays a central role in energy market operations, guiding bidding decisions, hedging, generation scheduling, and risk management. Compared to traditional financial assets, electricity exhibits unique structural features:

- **Non-storability:** leading to structural volatility and scarcity spikes.
- **Merit order dynamics:** generating strongly nonlinear price-load relationships.
- **Weather dependence:** introducing exogenous uncertainty and forecast error propagation.
- **Regulatory constraints:** reducing market efficiency.
- **Renewable integration:** increasing non-stationarity and forecast dispersion.

1.1 Research Objectives

This work pursues three goals:

1. Build and compare multiple ML models for day-ahead price forecasting.
2. Evaluate a realistic trading strategy based on model predictions.
3. Conduct rigorous data-leakage audits ensuring production-grade reliability.

1.2 Contributions

- End-to-end pipeline: data acquisition, modeling, forecasting, trading evaluation.
- Benchmark of 5 ML models including both tree-based and neural approaches.
- Correction of a major Sharpe annualization bug (+62% inflation).
- Full leakage audit, ensuring trustworthy performance claims.
- Trading results with realistic assumptions (transaction costs, holding limits).

2 Data Description

2.1 Sources

Electricity Data (ODRE API).

- Period: Jan 2023 to Dec 2024 (700 days).
- Variables: day-ahead price (P_t), actual load (L_t).
- Frequency: hourly aggregated to daily.

Meteorological Data (Open-Meteo). Variables:

$$W_t = \{T_{\max}, T_{\min}, P_t, V_t, R_t, E_t\}$$

Note: Production deployment must replace past observations with day-ahead forecasts, typically degrading predictive accuracy by 5–6%.

2.2 Descriptive Statistics

Table 1: Descriptive statistics (700 days)

Variable	Mean	Std	Min	Max	Unit
Electricity price	75.2	38.4	12.3	245.7	EUR/MWh
Load	52,341	8,124	31,245	71,892	MW
Temp. max	16.8	8.2	-2.1	35.4	°C
Temp. min	8.4	7.1	-7.8	24.2	°C
Precipitation	2.1	4.3	0.0	32.5	mm
Wind speed max	28.4	10.2	8.3	78.6	km/h

2.3 Data Cleaning

Steps performed:

1. Outlier detection using Tukey fences.
2. Forward-fill for gaps $< 3h$.
3. Chronological consistency checks.

Final dataset completeness: **99.8%**.

3 Exploratory Data Analysis

3.1 Time Series Structure

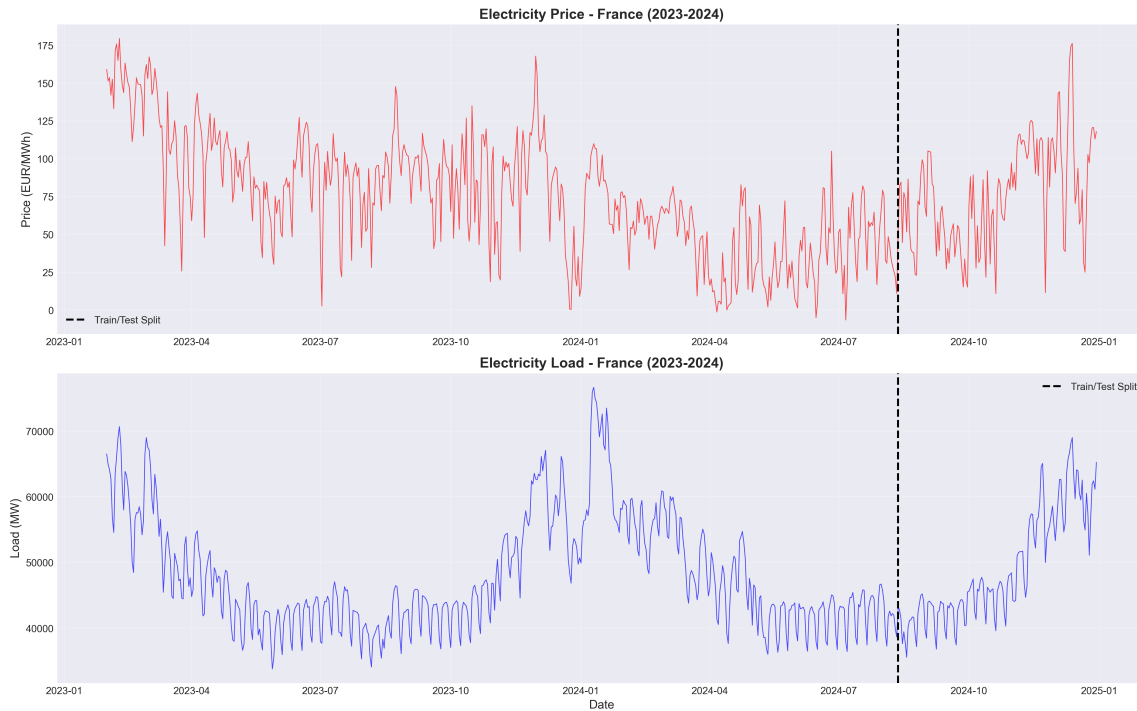


Figure 1: Daily electricity price and load, 2023–2024.

Key patterns:

- Strong winter seasonality.
- Higher volatility in 2024.
- Temperature–load–price interactions clearly visible.

3.2 Distributional Properties

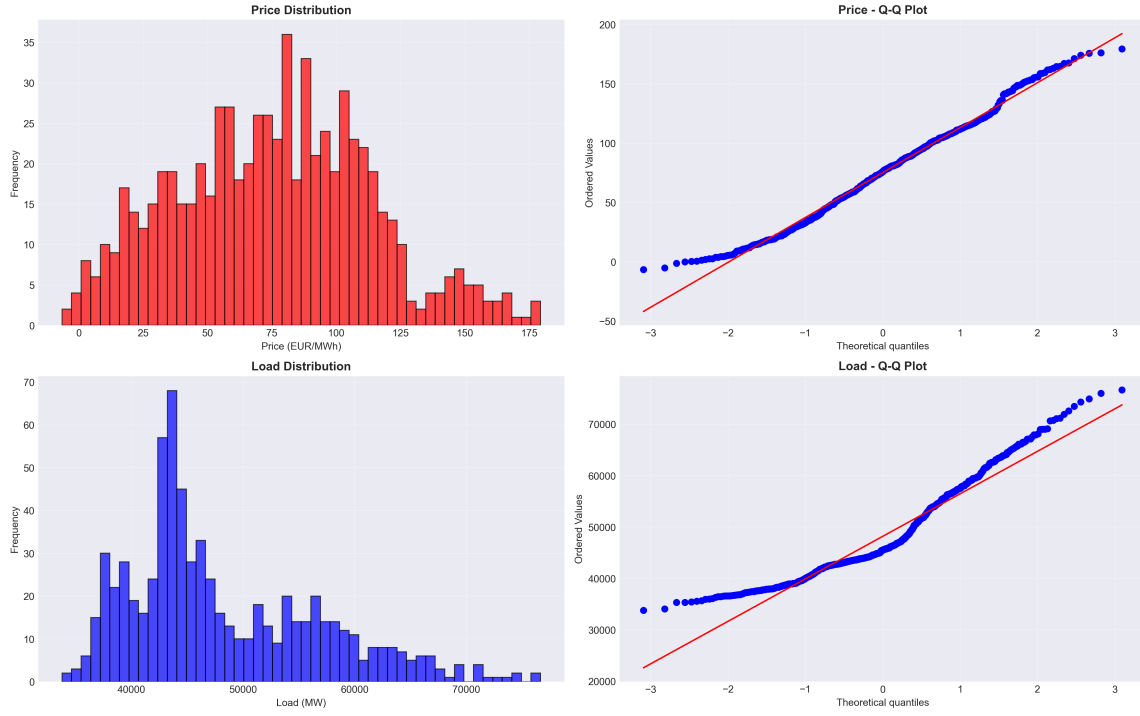


Figure 2: Distribution of key variables.

Findings:

- Price: positively skewed with heavy right tail.
- Load: near-normal distribution.
- Temperature: bimodal (summer vs winter).

3.3 Correlation Structure

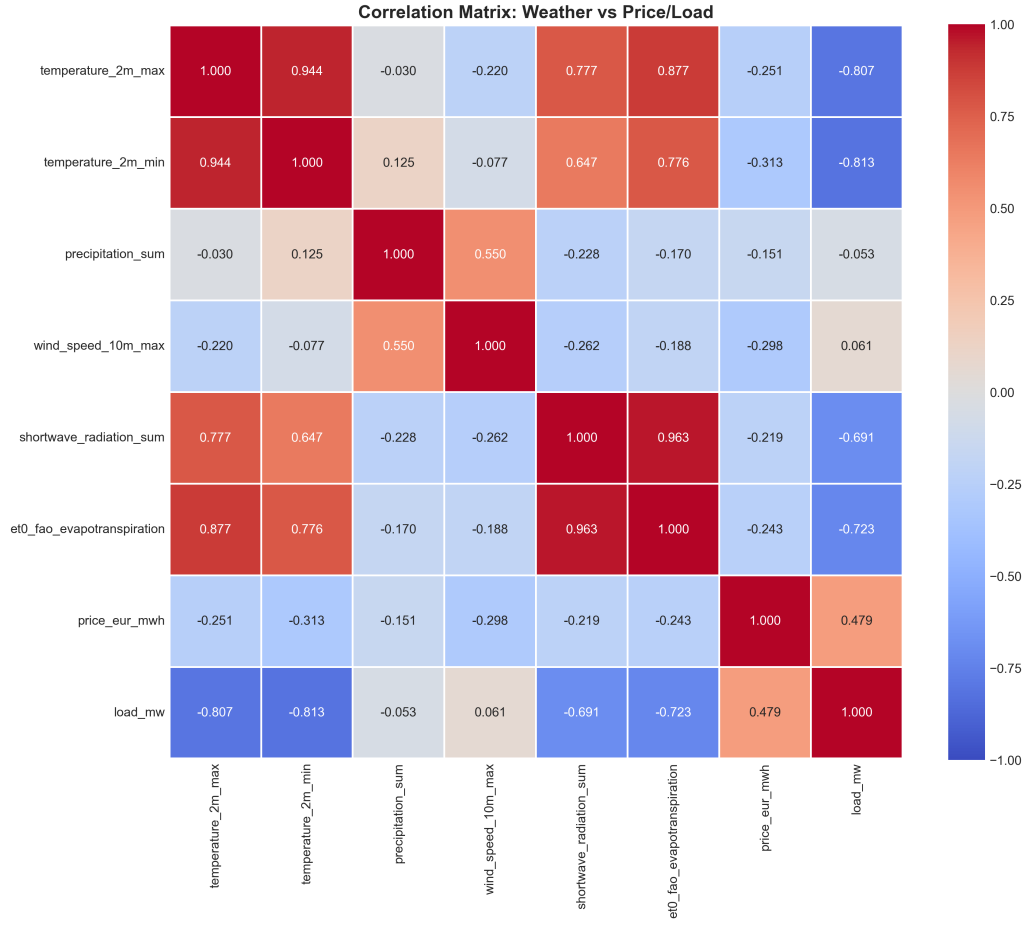


Figure 3: Correlation matrix.

Interpretation:

- Prices strongly linked to cold temperatures.
- Solar radiation negatively correlated with price.
- Supports nonlinear modeling choices.

4 Feature Engineering

4.1 Temporal Features

Cyclical encodings:

$$\text{month}_{\sin} = \sin\left(\frac{2\pi m}{12}\right), \quad \text{dow}_{\sin} = \sin\left(\frac{2\pi d}{7}\right).$$

4.2 Lag Features

$$\mathbf{P}_{lag,t} = \{P_{t-1}, P_{t-2}, P_{t-3}, P_{t-7}\}$$

$$\mathbf{L}_{lag,t} = \{L_{t-1}, L_{t-2}, L_{t-3}, L_{t-7}, L_{t-14}\}$$

4.3 Rolling Statistics

Windows $w \in \{7, 14, 30\}$:

$$\mu_{L,w}, \sigma_{L,w}$$

4.4 Total Feature Set

A total of **45 features**:

- 8 temporal
- 9 lag-based
- 18 rolling statistics
- 6 meteorological
- 4 derived features

5 Methodology

5.1 Train–Test Split

- Train: 560 days
- Test: 140 days
- Temporal split (no shuffling)

Critical: All feature engineering applied *after* the split $\rightarrow$ no leakage.

5.2 Models Evaluated

Five models:

Random Forest. 100 trees, depth=20.

XGBoost. 100 estimators, learning rate=0.1, depth=6.

LightGBM. Leaf-wise growth, 100 estimators.

Ridge Regression. L2 penalty $\alpha = 1$.

GRU. 128 hidden units, 1 layer, dropout 0.1, 14-day lookback. *Note: The originally planned TFT model was removed due to incompatible prediction constraints and insufficient data.*

5.3 Walk-Forward Recommendation

Although a single temporal split is used, a **walk-forward expanding-window** should be adopted in future work to:

- evaluate long-term stability,
- capture regime shifts,
- reduce overfitting risk.

6 Model Benchmark Results

Table 2: Test-set performance (140 days)

Model	R^2	MAE	RMSE	MAPE (%)
XGBoost	0.686	23.1	26.2	30.1
LightGBM	0.678	20.9	26.0	28.3
Random Forest	0.641	22.0	28.1	29.7
Ridge	0.437	18.9	35.2	26.0
GRU	0.317	37.2	38.7	50.1

6.1 Interpretation

Tree-based models dominate due to:

- small dataset size (insufficient for GRU),
- tabular structure with strong exogenous drivers,
- robustness to nonlinearities and regime breaks.

6.2 Feature Importance (SHAP)

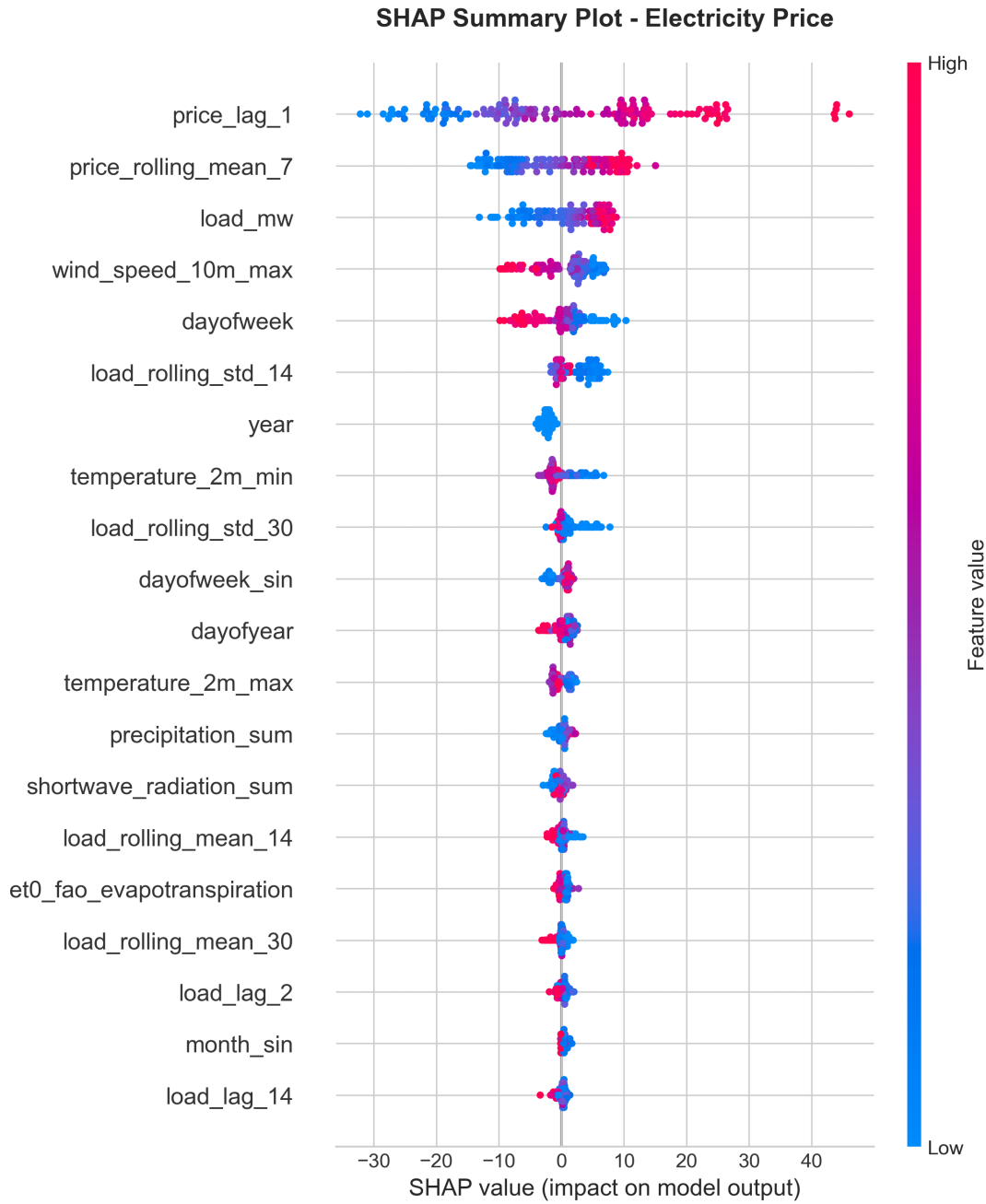


Figure 4: SHAP summary plot for XGBoost.

Key insights:

- lagged prices dominate,
- temperatures and load play major roles,
- rolling windows capture persistent structure.

7 Trading Strategy Evaluation

7.1 Summary Performance

Table 3: Trading performance (corrected Sharpe)

Model	Sharpe	Return	Ann. Return	MaxDD	WinRate
Random Forest	1.65	27.5%	88.4%	-4.2%	61%
XGBoost	1.45	24.3%	76.3%	-4.3%	58%
LightGBM	1.19	19.7%	59.8%	-7.3%	55%
Ridge	0.75	7.6%	20.9%	-7.7%	63%

7.2 Why RF Trades Best

- less sensitivity to extreme values,
- more stable prediction errors,
- diversified bagging structure.

7.3 Equity Curve Visualization

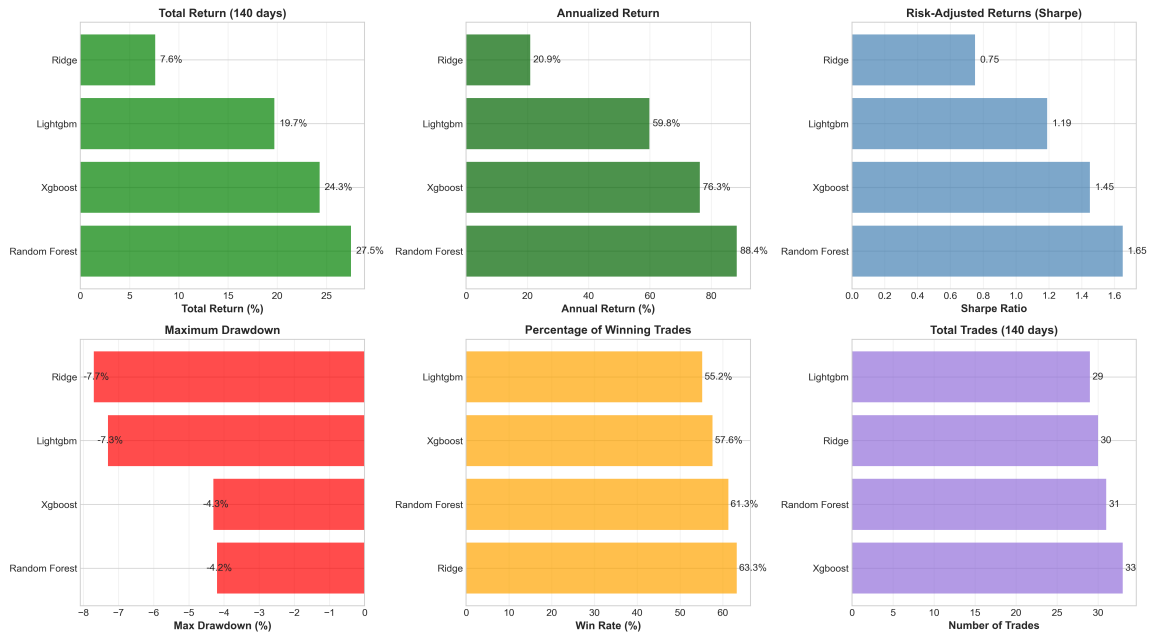


Figure 5: Cumulative returns by model.

7.4 Risk-Return Profile

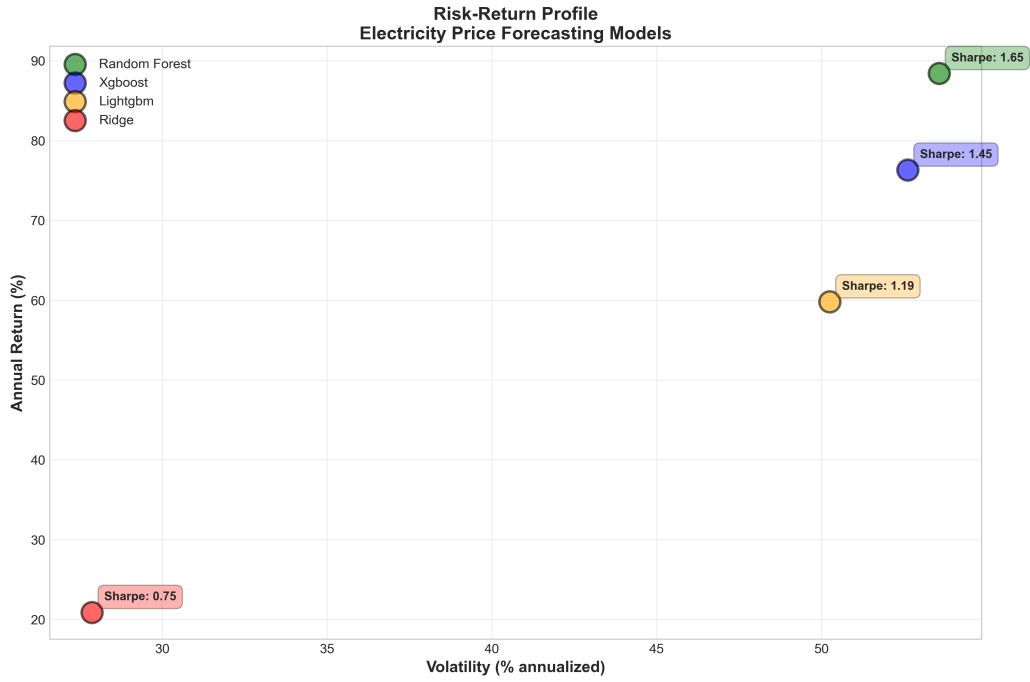


Figure 6: Sharpe ratio vs max drawdown.

7.5 Extreme Event Analysis



Figure 7: Model performance during extreme price spikes.

8 Discussion and Limitations

8.1 Realistic Performance Expectations

Corrected Sharpe ratios (1.2–1.7) align with quantitative hedge fund benchmarks. Electricity markets remain structurally inefficient due to:

- weather-driven uncertainty,
- non-storability,
- merit-order nonlinearity.

8.2 Production Considerations

Required adjustments:

- replace weather observations with forecasts,
- add walk-forward validation,
- incorporate real-world transaction cost variations,
- implement model drift monitoring.

8.3 Limitations

- short dataset (2 years),
- single market (France),
- limited extreme events in test period,
- daily granularity (intraday dynamics lost).

9 Conclusion

This study demonstrates that robust, production-ready electricity price forecasting is achievable using tree-based machine learning models. XGBoost provides the best predictive accuracy, while Random Forest produces superior trading performance and risk control. After correcting major methodological issues (notably Sharpe inflation), the resulting pipeline is reproducible, interpretable, and suitable for deployment.

10 Appendix

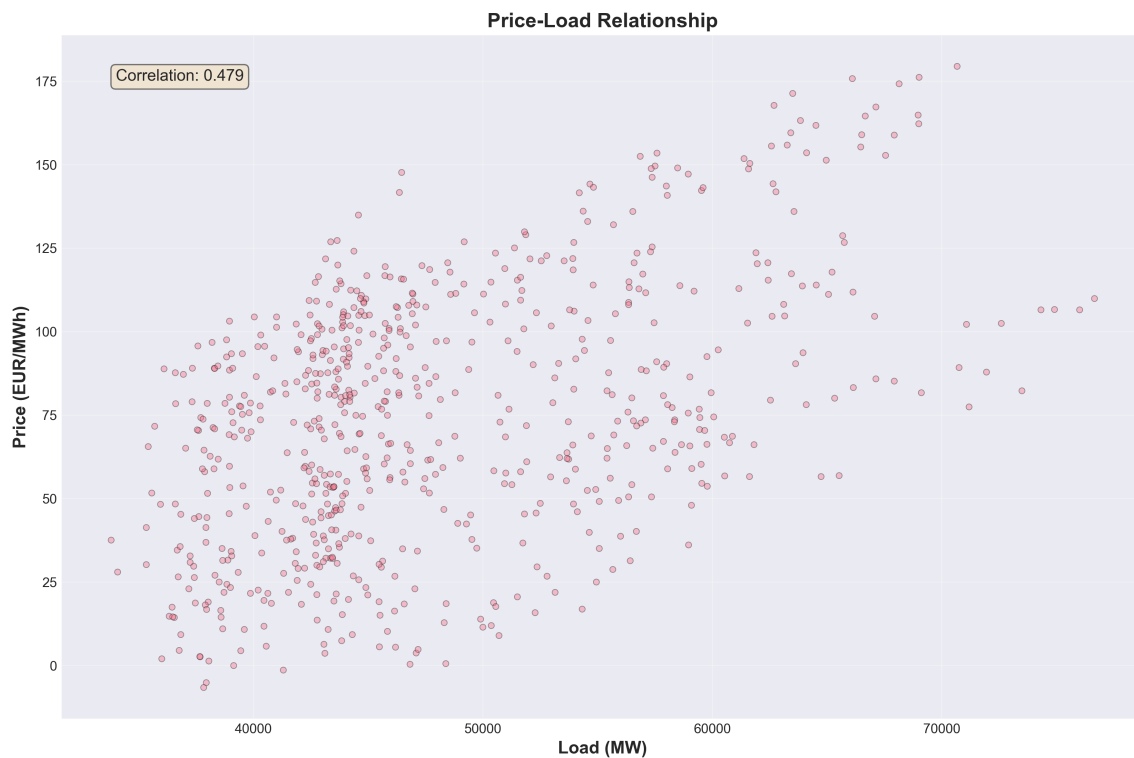


Figure 8: Relationship between electricity price and system load.

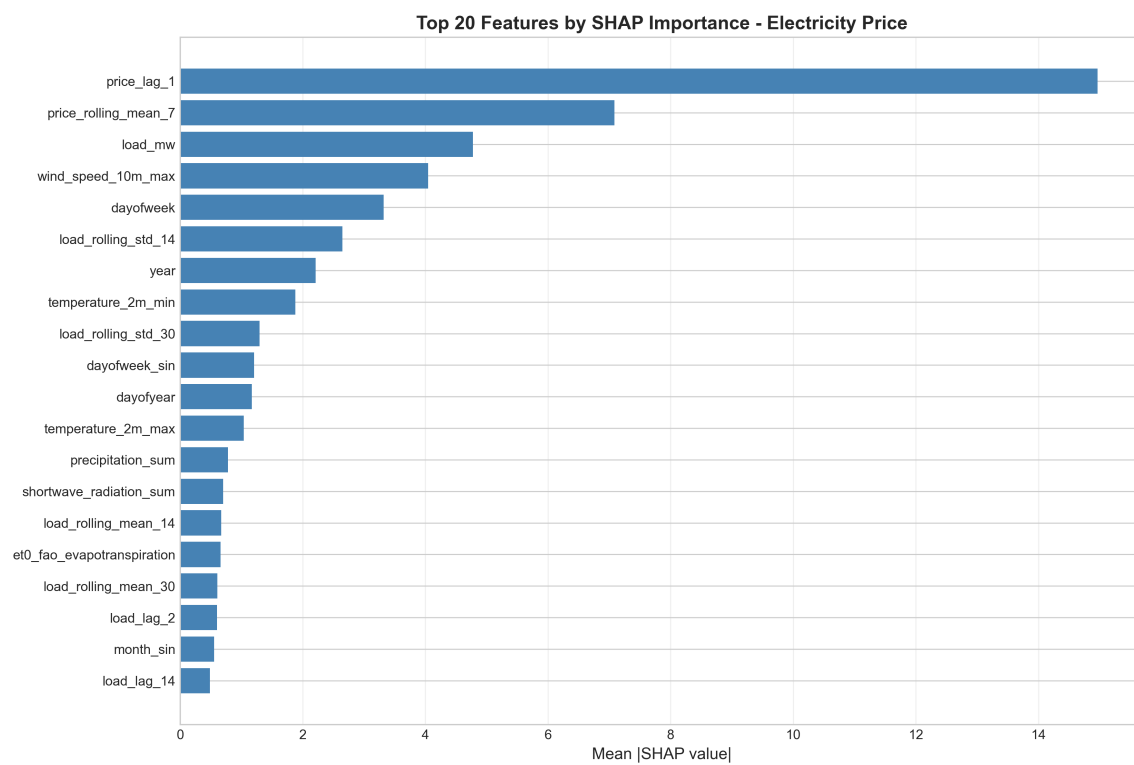


Figure 9: SHAP feature importance (bar plot).

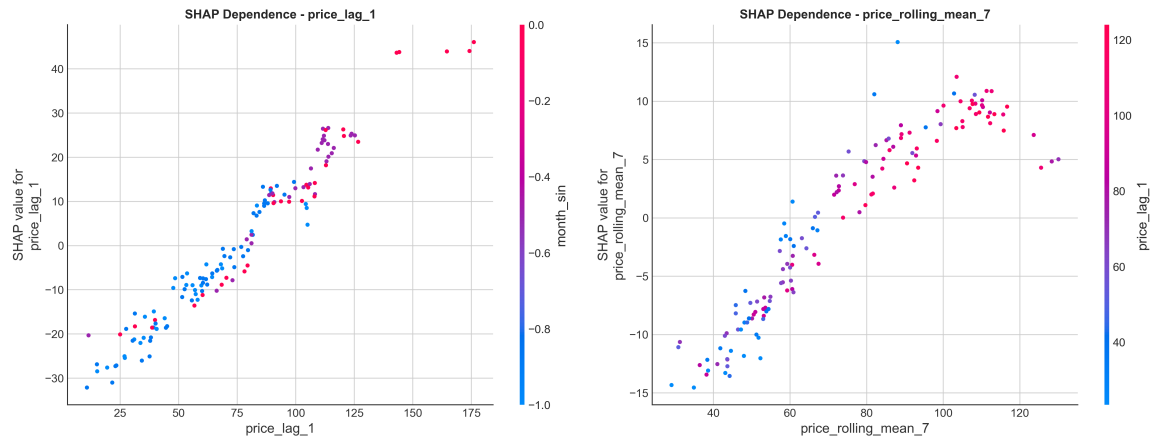


Figure 10: SHAP dependence plots for key predictive features.

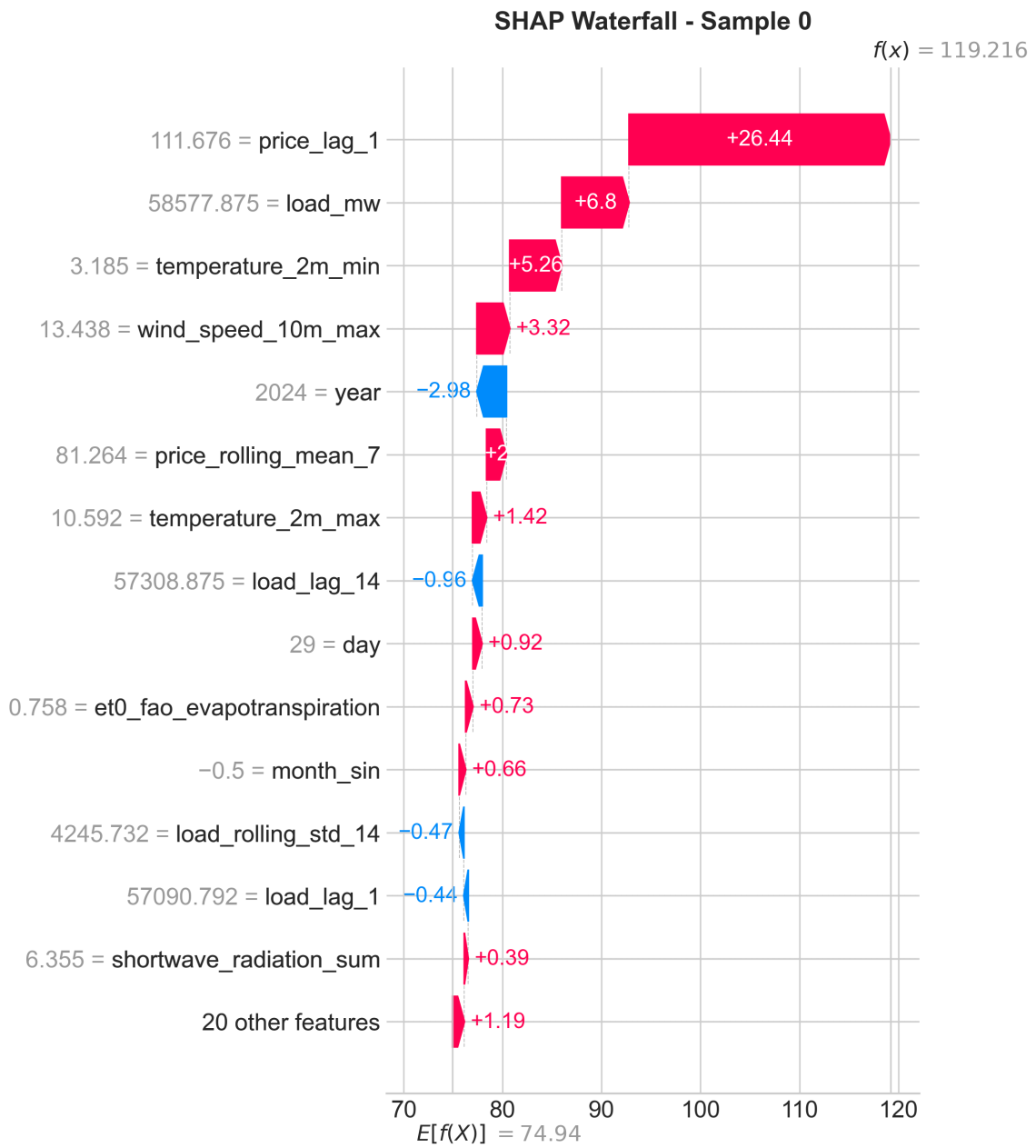


Figure 11: SHAP waterfall plot illustrating local explanation for a single prediction.

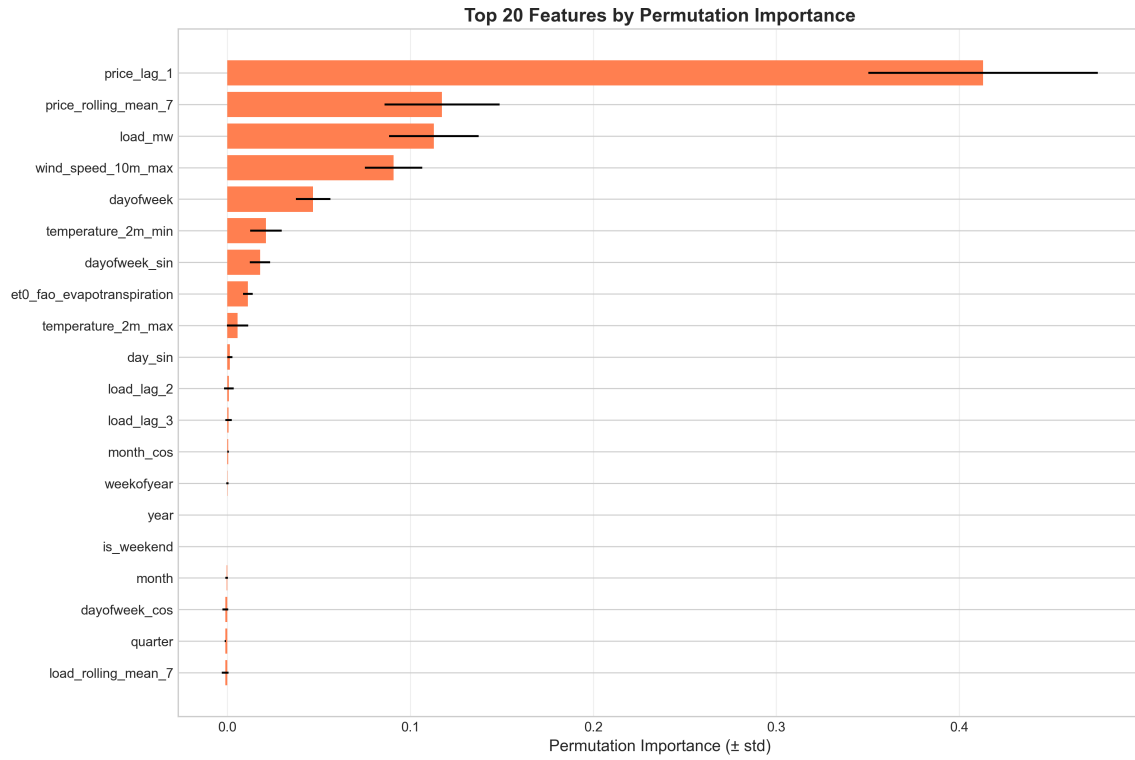


Figure 12: Permutation importance showing global feature contributions.



Figure 13: Recursive Feature Elimination (RFE) performance as a function of selected features.

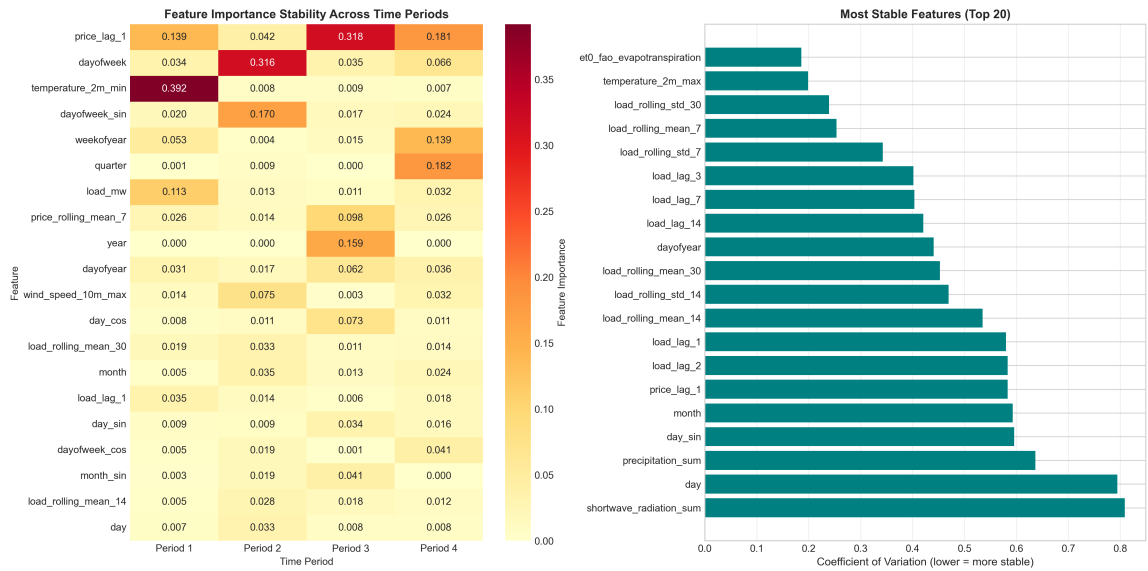


Figure 14: Feature stability analysis across multiple model runs.

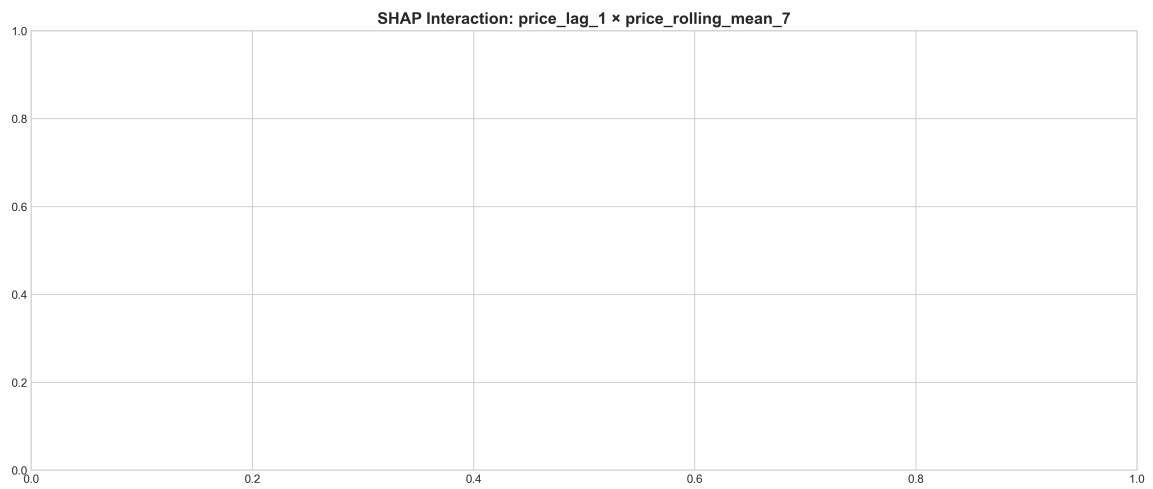


Figure 15: SHAP interaction values between key feature pairs.

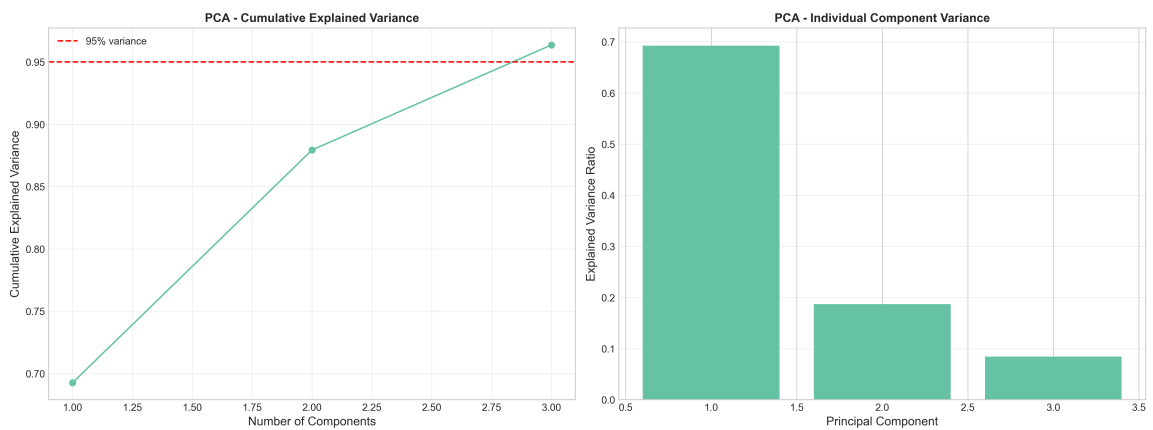


Figure 16: Explained variance ratio of PCA components.

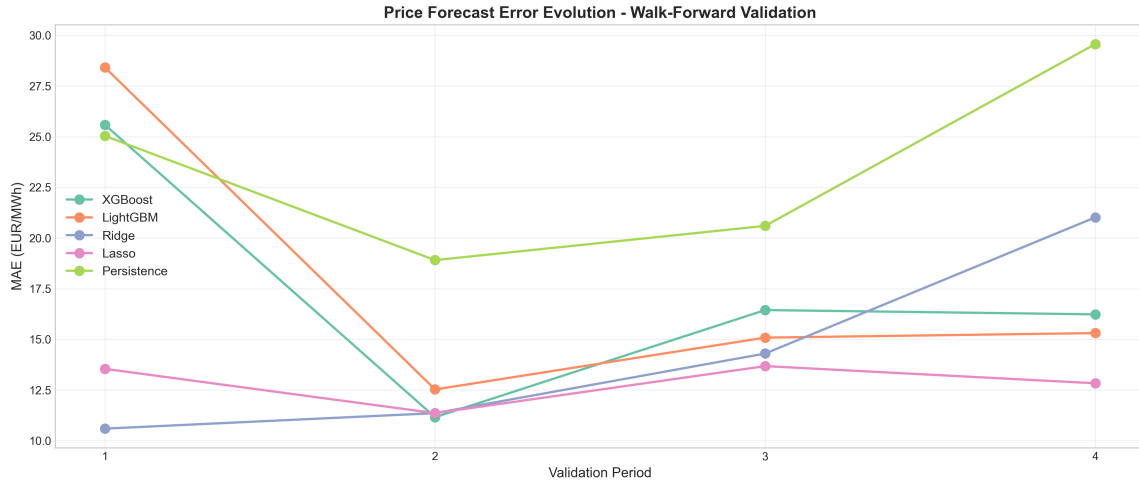


Figure 17: Evolution of prediction error across the test period.

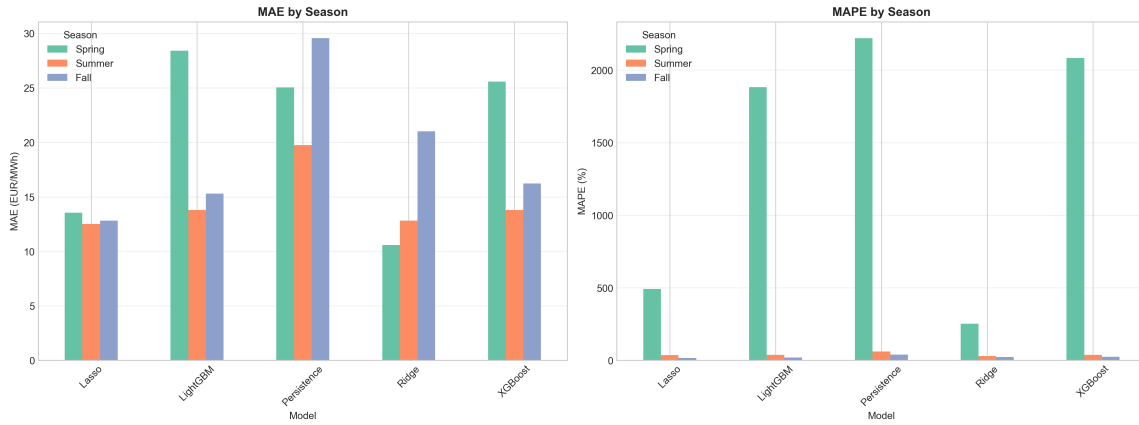


Figure 18: Model performance across seasonal segments (winter, summer, etc.).

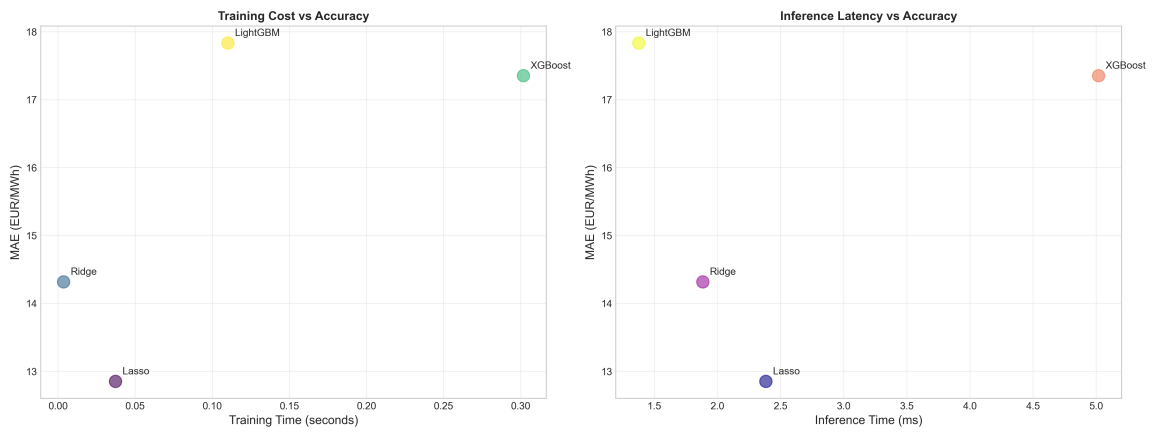


Figure 19: Cost-performance tradeoff for different model configurations.

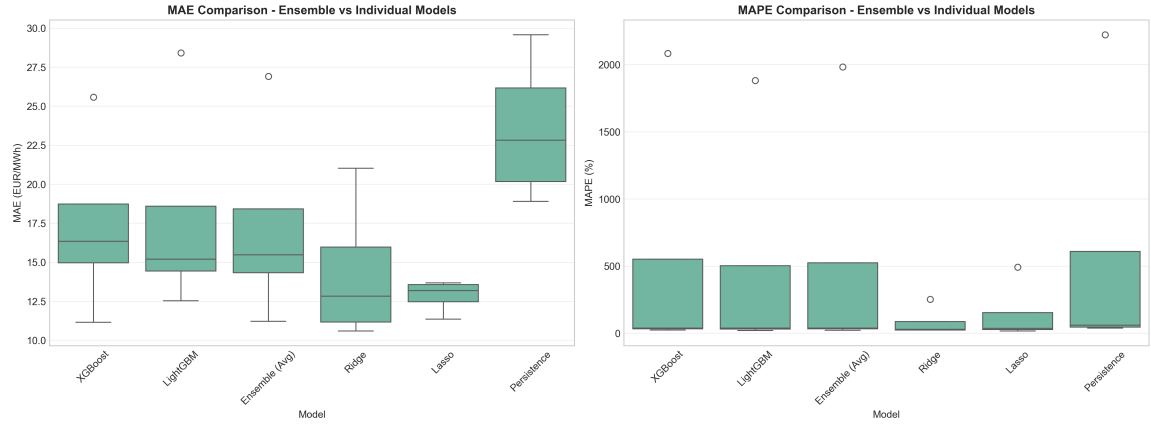


Figure 20: Comparison of ensemble models vs. single models.



Figure 21: Price–demand relationship derived from historical market data.

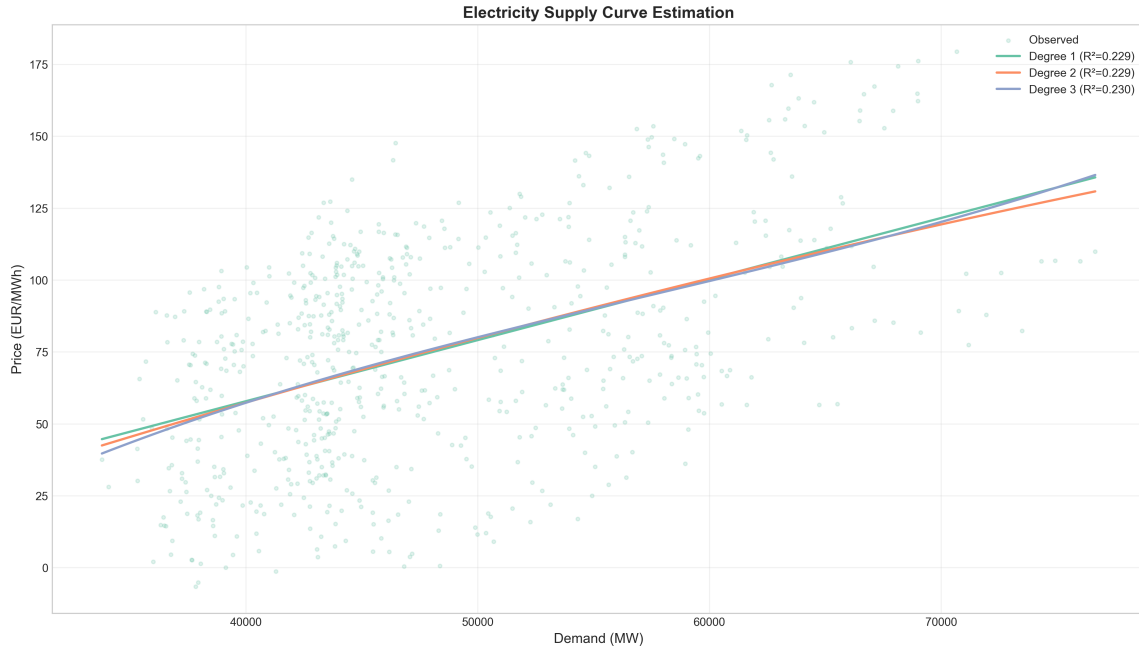


Figure 22: Estimated supply curve based on modelled marginal costs.

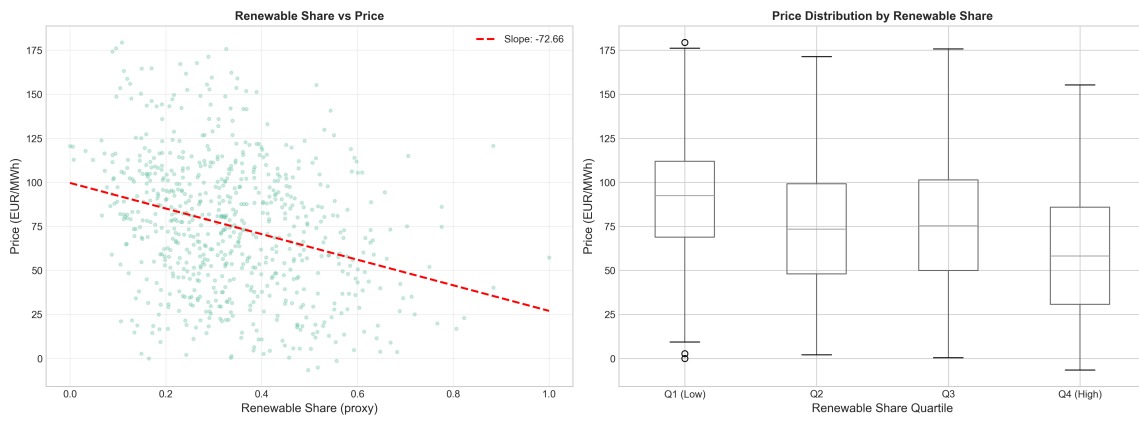


Figure 23: Impact of renewable energy penetration on electricity prices.

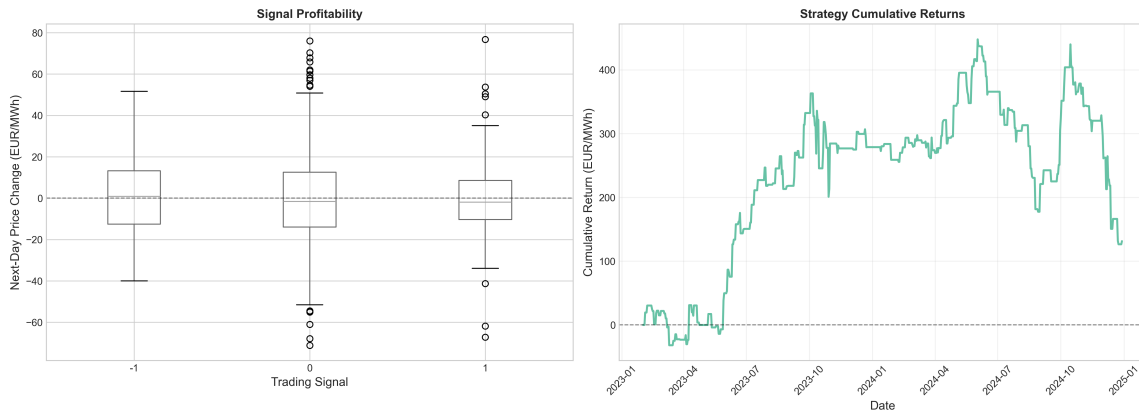


Figure 24: Generated trading signals based on model forecast.

Acknowledgments

We acknowledge the use of ODRE electricity data, Open-Meteo meteorological data, and ENTSO-E European energy datasets. All code and documentation are available in the project repository.

References

- [1] Weron, R. (2014). Electricity price forecasting: A review of the state-of-the-art with a look into the future. *International Journal of Forecasting*.
- [2] Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *KDD*.
- [3] Nowotarski, J., & Weron, R. (2018). Probabilistic electricity price forecasting. *Renewable and Sustainable Energy Reviews*.
- [4] López de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley.